

FOREST COVER TYPE

PREDICTION WITH XGBOOST

**ANALYZING ENVIRONMENTAL FACTORS FOR ECOLOGICAL
MONITORING**

AGENDA

- 1 PROJECT OVERVIEW
- 2 METHODOLOGY
- 3 RESULTS AND ANALYSIS
- 4 CONCLUSIONS
- 5 APPENDIX

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PROJECT OVERVIEW

RESUME

Develop a high-performance machine learning model capable of predicting the type of forest cover (1 to 7) based on geographical and cartographic variables.

Leveraging XGBoost, one of the most powerful algorithms for structured data classification.

BACKGROUND

Natural resource management often lacks precise knowledge of forest ecosystems across vast, hard-to-reach areas.

Traditional surveys are expensive and slow. This model utilizes 54 predictors (elevation, slope, soil type) from the US Forest Service.

IMPORTANCE

- Conservation: Identify forest zones for protection.
- Fire Prevention: Predict high-risk fuel types.
- Land-use: Data-driven government planning.
- Remote Sensing: Scale monitoring without physical site visits.

OBJECTIVES

- Accurately classify all 7 forest cover types (Spruce/Fir, Lodgepole Pine, Ponderosa Pine, Cottonwood/Willow, Aspen, Douglas-fir, Krummholz).
- Evaluate topographic vs. distance features.
- Learn complex ecological patterns.
- Achieve high predictive accuracy (XGBoost).

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METHODOLOGY

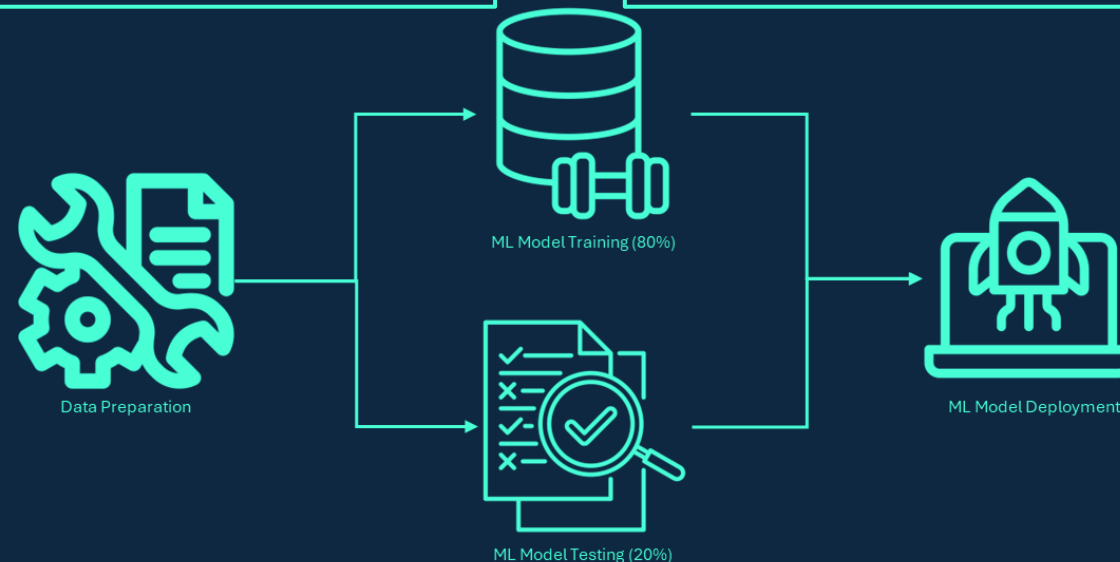
PREDICTOR VARIABLES

Cartographic Data

- Topography (0-2): Elevation, Aspect, and Slope.
- Distances and infrastructure (3-5): To Hydrology (Horiz/Vert), Roadways, and Fire Points.
- Hillshade (6-8): solar lighting at 9:00 a.m., 12:00 p.m., and 3:00 p.m., respectively.
- Horizontal Distance to Fire Points (9).
- Wilderness Areas (10–13): 4 binary ecological zones.
- Soil Types (14-53): 40 binary categories (USFS complexes).

DATA PREPARATION & PIPELINE

- Training Architecture: Processing a substantial dataset of 581,012 samples to feed the boosting architecture efficiently:
 - Stratified 80/20 Split:
 - Train: 464,809 rows.
 - Test: 116,203 rows.
 - Preserves target representation across unbalances.
 - Target Alignment: Converted target range from (1–7) to (0–6) to adhere to XGBoost strict index mapping.
- XGBoost Parameters:
 - Algorithm: XGBClassifier
 - n_estimators: 150
 - max_depth: 6
 - learning_rate: 0.1
 - subsample: 0.8
 - eval_metric: mlogloss



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RESULTS AND ANALYSIS

MODEL PERFORMANCE EVALUATION

83.08%

Global Accuracy

Evaluated on 116,203 test samples

Class (Cover Type)	Precision	Recall	F1-Score
1: Spruce/Fir	0.82	0.80	0.81
2: Lodgepole Pine	0.83	0.87	0.85
3: Ponderosa Pine	0.84	0.88	0.86
4: Cottonwood/Willow	0.85	0.82	0.83
5: Aspen	0.86	0.43	0.57
6: Douglas-fir	0.80	0.68	0.74
7: Krummholz	0.92	0.86	0.89

KEY STRATEGIC INSIGHTS

- Dominant Ecosystems Stability: The model achieves robust F1-Scores between 0.81 and 0.85 for Spruce/Fir and Lodgepole Pine. This proves XGBoost successfully learned to navigate and classify the vast majority of the forest layout despite the heavy class imbalance.
- High-Altitude Precision: Class 7 (Krummholz) reaches an outstanding 0.92 Precision score. The model rarely triggers a false alarm for this class because it has perfectly integrated the extreme elevation rules where this species lives.

CONFUSION MATRIX & CLASS INSIGHTS

Real \ Pred	C1	C2	C3	C4	C5	C6	C7
C1 (Spruce)	33742	8316	4	0	18	7	281
C2 (Lodgepole)	6596	49353	363	0	108	208	33
C3 (Ponderosa)	0	447	6324	56	1	323	0
C4 (Cottonwood)	0	0	79	448	0	22	0
C5 (Aspen)	11	1028	35	0	813	12	0
C6 (Douglas)	6	345	743	26	2	2351	0
C7 (Krummholz)	576	14	0	0	0	0	3512

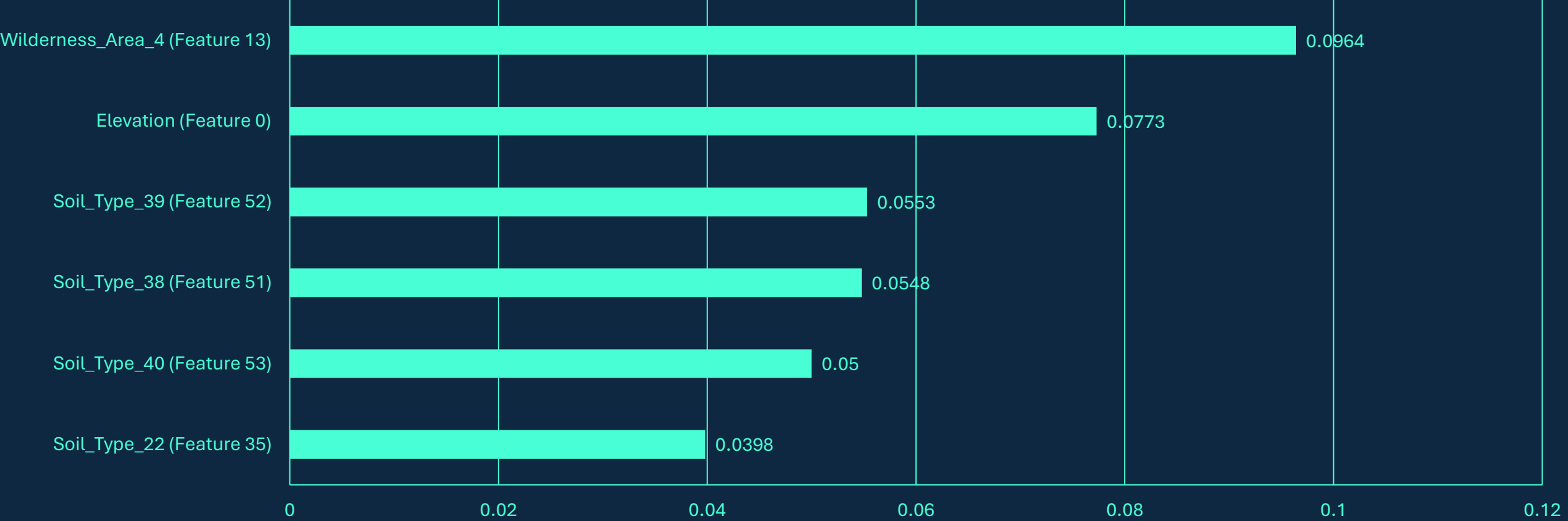
KEY STRATEGIC INSIGHTS

- Dominant Class Overlap: The most significant confusion occurs between Class 1 (Spruce/Fir) and Class 2 (Lodgepole Pine). This aligns with nature: both share high elevations and subalpine climates.
- The Class 5 (Aspen) Challenge: Class 5 yielded a low recall (0.43) due to 1,028 actual patches being misclassified as Class 2. Aspen acts as a temporary/transitional species in areas dominated by Lodgepole Pines, creating a complex botanical mix.

RESULTS AND ANALYSIS

TOP 10 FEATURE IMPORTANCE (XGBOOST WEIGHT)

The model attributes high discriminative power to engineered vectors and primary topographic metrics:



Feature 13 (Cache la Poudre Wilderness Area) and Feature 0 (Elevation) stand as the strongest biophysical constraints for tree layout distribution.

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CONCLUSIONS

Robust Predictive Stability in Dominant Ecosystems

Ecosystem Learning: The model demonstrated high stability and robustness when classifying the dominant forest species, achieving an F1-Score of 0.81 for Spruce/Fir (Class 1) and 0.85 for Lodgepole Pine (Class 2). This confirms that XGBoost successfully mitigated the dataset's severe class imbalance (which heavily concentrates on these two categories) without relying on destructive synthetic oversampling techniques.

Core Biophysical Drivers of Forest Layout

Primary Environmental Filters: Feature importance analysis revealed that forest distribution is far from random; instead, it is strictly governed by two primary factors: Wilderness_Area_4 / Cache la Poudre (Feature 13) with a weight of 0.0964, and Elevation (Feature 0) with 0.0773. These act as the first and most rigorous biological exclusion filters within the ecosystem.

Strict Niche Specialization at High Altitudes

Krummholz Precision: The extreme-climate vegetation Krummholz (Class 7) achieved the highest precision level across the entire system with an outstanding score of 0.92. This result provides strong evidence that the algorithm perfectly mapped the restrictive altitudinal and climatic rules that define this ecological niche, drastically minimizing the rate of false alarms in summit zones.

Micro-Geological and Human Reshaping Factors

Socio-Environmental Impact: The model's success was consolidated by the predictive weight of high-altitude alpine soil complexes (Soil_Types 38, 39, and 40 / Features 51, 52, and 53), which serve as definitive geological boundaries. Additionally, variables such as solar illumination (Hillshade variables) and Horizontal_Distance_To_Roadways (Feature 5) provided the necessary context to evaluate how terrain moisture retention and human-induced landscape fragmentation actively alter natural forest distribution.

The Subalpine Territorial Overlap Challenge

The Aspen and Spruce Bias: The most significant predictive challenge was identified within Aspen (Class 5), which yielded a critical Recall of just 0.43 because 1,028 of its actual zones were misclassified as Lodgepole Pine. Furthermore, over 14,900 cross-misclassifications occurred between Spruce/Fir and Lodgepole Pine. These metrics accurately reflect biological reality: intense territorial competition and diffuse transitions along the altitudinal borders of these species.

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Index	Feature Name	Description / Ecological Meaning
0	Elevation	Elevation in meters above sea level
1	Aspect	Aspect in degrees azimuth (compass direction of slope)
2	Slope	Slope tilt inclination in degrees (0° to 90°)
3	Horizontal_Distance_To_Hydrology	Horizontal distance to nearest surface water features
4	Vertical_Distance_To_Hydrology	Vertical distance to nearest surface water features
5	Horizontal_Distance_To_Roadways	Horizontal distance to nearest roadway infrastructure
6	Hillshade_9am	Hillshade index at 9 a.m., summer solstice (0 to 255)
7	Hillshade_Noon	Hillshade index at noon, summer solstice (0 to 255)
8	Hillshade_3pm	Hillshade index at 3 p.m., summer solstice (0 to 255)
9	Horizontal_Distance_To_Fire_Points	Horizontal distance to nearest historical wildfire ignition points
10	Wilderness_Area_1	Rawah Wilderness Area (Binary: 0 or 1)
11	Wilderness_Area_2	Neota Wilderness Area (Binary: 0 or 1)
12	Wilderness_Area_3	Comanche Peak Wilderness Area (Binary: 0 or 1)
13	Wilderness_Area_4	Cache la Poudre Wilderness Area (Binary: 0 or 1)
14	Soil_Type_1	Cathedral family - Rock outcrop complex, extremely stony
15	Soil_Type_2	Vanet - Ratake families complex, very stony
16	Soil_Type_3	Haploborolls - Rock outcrop complex, rubbly
17	Soil_Type_4	Ratake family - Rock outcrop complex, rubbly
18	Soil_Type_5	Vanet family - Rock outcrop complex, rubbly
19	Soil_Type_6	Vanet - Wetmore families - Rock outcrop complex, stony
20	Soil_Type_7	Gothic family, stony loam
21	Soil_Type_8	Supervisor - Limber families complex, stony loam
22	Soil_Type_9	Troutville family, stony loam
23	Soil_Type_10	Bullwark - Catamount families - Rock outcrop complex, rubbly
24	Soil_Type_11	Bullwark - Catamount families - Rock land complex, rubbly
25	Soil_Type_12	Legault family - Rock outcrop complex, stony
26	Soil_Type_13	Catamount family - Rock outcrop - Rubble land complex, stony
27	Soil_Type_14	Pachic Argiborolls - Aquolls complex, silt loam
28	Soil_Type_15	Unspecified in USFS records (unpopulated)
29	Soil_Type_16	Cryaquolls - Cryoborolls complex, plains/valleys
30	Soil_Type_17	Gateview family, heavy stony loam

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Index	Feature Name	Description / Ecological Meaning
31	Soil_Type_18	Rogert family, very stony loam
32	Soil_Type_19	Typic Cryaquolls - Borohemists complex, highly organic
33	Soil_Type_20	Typic Cryaquepts - Typic Cryaquolls complex, wet soils
34	Soil_Type_21	Typic Cryaquolls - Leighcan family, till substratum
35	Soil_Type_22	Leighcan family, till substratum, extremely stony
36	Soil_Type_23	Leighcan family, till substratum, stony loam
37	Soil_Type_24	Leighcan family, warm substratum, extremely stony
38	Soil_Type_25	Granile - Catamount families complex, very stony loam
39	Soil_Type_26	Leighcan family, warm substratum, stony loam
40	Soil_Type_27	Leighcan family, stony loam (lower drainage zones)
41	Soil_Type_28	Leighcan family, extremely stony loam (transition zone)
42	Soil_Type_29	Como family, extremely stony loam
43	Soil_Type_30	Como - Legault families complex, extremely stony loam
44	Soil_Type_31	Leighcan - Catamount families complex, extremely stony loam
45	Soil_Type_32	Catamount family - Rock outcrop - Rubble land complex, rubbly
46	Soil_Type_33	Leighcan - Catamount families - Rock outcrop complex, rubbly
47	Soil_Type_34	Cryorthents - Rock land complex, alpine/barren
48	Soil_Type_35	Cryumbrepts - Rock outcrop - Cryorthents complex, high-altitude
49	Soil_Type_36	Bross family - Rock outcrop complex, alpine ridge soil
50	Soil_Type_37	Rock outcrop - Cryumbrepts - Cryorthents complex, alpine ridge
51	Soil_Type_38	Leighcan - Catamount families complex, alpine, extremely stony
52	Soil_Type_39	Leighcan - Moran families complex, deep glacial till
53	Soil_Type_40	Moran family - Cryaquolls - Rock outcrop complex, high alpine